

ERA5 reanalysis

1940–2024 · 6-hourly · 0.25° · CDS



Moisture transport & dynamics

IVT, IWV, θ_e , PV (850 / 500 / 250 hPa)



Atmospheric-river detection

Guan–Waliser · period-internal 85th-pct IVT



Event extraction

intensity · footprint · duration · landfall



Three-window contrast

intensity trajectory → Fig. 2, 4

Thermo / dynamic split

$\text{IVT} = \text{IWV} \cdot \hat{V} \rightarrow \text{Fig. 3}$